

There are a total of five problems. You have to solve all the problems.

Problem 1 (CO1): DFA and Regular Languages (15 points)

Let $\Sigma = \{a, b\}$. Consider the following languages over Σ .

$$\begin{aligned} L_1 &= \{w : \text{length of } w \text{ is two more than multiple of four}\} \\ L_2 &= \{w : \text{every even position letter in } w \text{ is different from the first letter of } w\} \\ L_3 &= \{w : \text{every } 2k + 1 \text{ position in } w \text{ is } b, \text{ where } k \geq 0\} \\ L_4 &= \{w : \text{every } 2k + 1 \text{ position in } w \text{ is } a, \text{ where } k \geq 0\} \end{aligned}$$

Now solve the following problems.

- Give the state diagram for a DFA that recognizes L_1 . (3 points)
- Give the state diagram for a DFA that recognizes L_2 . (3 points)
- Give the state diagram for a DFA that recognizes L_3 . (3 points)
- If you were to use the “cross product” construction to obtain a DFA for the language $L_2 \cap (L_3 \cup L_4)$, how many states would it have? (1 point)
- Find all four-letter strings in $L_2 \cap (L_3 \cup L_4)$. (1 point)
- Give the state diagram for a DFA that recognizes $L_2 \cap (L_3 \cup L_4)$ using only four states. (2 points)
- Find a four-letter string in $\overline{L_3} \circ L_4$. [Recall: $\overline{L}$ denotes the complement of the language L i.e., $\overline{L} = \Sigma^* - L$] (1 point)
- Is $\overline{L_3} \circ L_4 = \overline{L_3}$? Give justification for your answer. (1 point)

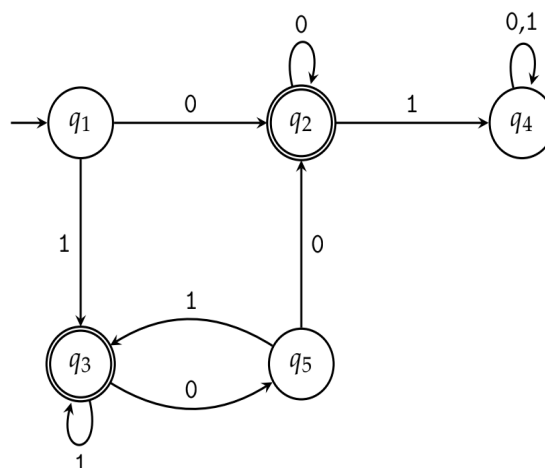
Problem 2 (CO2): Converting Regular Expressions to NFA (10 points)

Convert the following regular expression over $\Sigma = \{a, b, c\}$ into an equivalent NFA. Note that $R_1 + R_2$ is the same as $R_1 \cup R_2$.

$$((ab + c)^* + bc^*)ac$$

Problem 3 (CO2): Converting Finite Automata to Regular Expressions (10 points)

Convert the following finite automata into an equivalent regular expression using the state elimination method. You must eliminate q_2 first, then q_4 , then q_3 , and finally q_5 . Show each step of the process.



Problem 4 (CO1): Regular Expressions (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

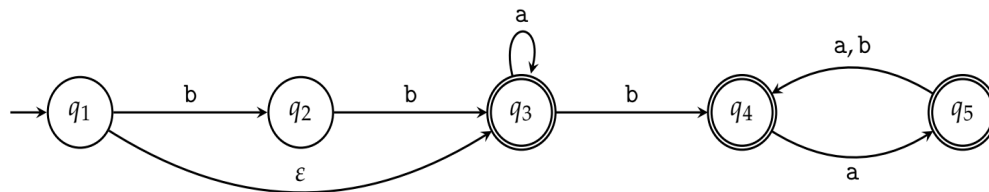
$$\begin{aligned} L_1 &= \{w : \text{length of } w \text{ is exactly } 3\} \\ L_2 &= \{w : \text{the third last digit of } w \text{ is } 1\} \\ L_3 &= \{w : w \text{ contains at most two } 00\} \\ L_4 &= \overline{L_1^*} \cap \overline{L_2} \end{aligned}$$

Now solve the following problems.

- Give a regular expression for the language L_1 . (1 point)
- Give a regular expression for the language L_1^* . (1 point)
- Give a regular expression for the language $\overline{L_1^*}$. [Recall: $\overline{L}$ denotes the complement of the language L i.e., $\overline{L} = \Sigma^* - L$] (2 points)
- Give a regular expression for the language L_2 . (2 points)
- Give a regular expression for the language $\overline{L_3}$. (2 points)
- Give a regular expression for the language L_4 . (2 points)

Problem 5 (CO2): Subset Construction Method (5 points)

Consider the following NFA:



Now answer the following questions. [Note: You do not need to convert the given NFA into its equivalent DFA to answer the questions.]

- If you convert the given NFA into an equivalent DFA using the subset construction method, what is the maximum number of states that the DFA can have? (1 point)
- what is the maximum number of accepting states that the equivalent DFA can have? (1 point)
- Write the ϵ -closure of state q_1 in the given NFA. (1 point)
- Write the subset of states of the given NFA that will be the starting state in its equivalent DFA. (1 point)
- What is $\delta(\{q_1, q_2\}, b)$ in the given NFA? [Recall: $\delta(\{q\}, a)$ is the set of states in which the NFA transitions when it is in state q and receives input a .] (1 point)

After you have finished the exam, please indicate where you stand on the smiley-face spectrum.

